Governors State University
OPUS Open Portal to University Scholarship

All Capstone Projects | Student Capstone Projects

Fall 2023

Cure Unity

Saisandesh Devireddy
Governors State University

Follow this and additional works at: https://opus.govst.edu/capstones

Part of the Computer Sciences Commons

Recommended Citation

Devireddy, Saisandesh, "Cure Unity" (2023). *All Capstone Projects*. 672.
https://opus.govst.edu/capstones/672

For more information about the academic degree, extended learning, and certificate programs of Governors State University, go to http://www.govst.edu/Academics/Degree_Programs_and_Certifications/

Visit the Governors State Computer Science Department
This Capstone Project is brought to you for free and open access by the Student Capstone Projects at OPUS Open Portal to University Scholarship. It has been accepted for inclusion in All Capstone Projects by an authorized administrator of OPUS Open Portal to University Scholarship. For more information, please contact opus@govst.edu.

CURE UNITY

By

Sai Sandesh, DEVIREDDY
B.Tech, JNTUA COLLEGE OF ENGINEERING, 2019

GRADUATE CAPSTONE SEMINAR PROJECT

Submitted in partial fulfillment of the requirements

For the Degree of Master of Science,

With a Major in Computer Science

Governors State University
University Park, IL 60484

2023

ABSTRACT

In response to the rising demand for improved healthcare services, our research introduces the Medicine Donation System—an innovative web application. This platform is designed to enhance the accessibility of medicines for those in need. It offers user-friendly features like registration, medicine donation, NGO integration, robust monitoring and control, inventory management, streamlined medicine requests, insightful report generation, and real-time chatbot support.

Our system streamlines the medicine donation process, making it easy for users to register and contribute. NGOs can seamlessly integrate with the platform to receive donations, while advanced monitoring and control mechanisms ensure the safety and effectiveness of medicines. Users can request medicines effortlessly, and the system provides valuable insights into donation trends through comprehensive reports. With a live chatbot for prompt assistance, our web application aims to revolutionize medical aid provision.

Table of Content

1 Project Description 1
1.1 Competitive Information 1
1.2 Relationship to Other Applications/Projects 2
1.3 Assumptions and Dependencies 2
1.4 Future Enhancements 4
1.5 Definitions and Acronyms 4
2 Technical Description 5
2.1 Project/Application Architecture 5
2.2 Project/Application Information flows 6
2.3 Interactions with other Projects (if Any) 10
2.4 Interactions with other Applications 10
2.5 Capabilities 11
2.6 Risk Assessment and Management 12
3 Project Requirements 13
3.1 Identification of Requirements 13
3.2 Operations, Administration, Maintenance and Provisioning (OAM&P) 14
3.3 Security and Fraud Prevention 15
3.4 Release and Transition Plan 15
4 Project Design Description 17
5 Project Internal/external Interface Impacts and Specification 18
6 Project Design Units Impacts 18
6.1 Functional Area/Design Unit A 18
6.1.1 Functional Overview 19
6.1.2 Impacts 19
6.1.3 Requirements 19
6.2 Functional Area/Design Unit B 20
6.2.1 Functional Overview 20
6.2.2 Impacts 20
6.2.3 Requirements 20
7 Open Issues 21
8 Acknowledgements 21
9 References 21
10 Appendices 21

1 Project Description

Cureunity is a vital initiative aimed at addressing the pressing issue of limited access to essential medicines, particularly for those who cannot afford them. This project employs a modern and user-friendly approach by utilizing the MVC Core framework with a SQL Server database, ensuring a seamless and efficient experience for both users and administrators.

Importance of the Project:

Cureunity emerges as a crucial response to the disparity in healthcare access, specifically in the realm of medication. By providing a platform where individuals can order medicines for free, the project strives to contribute to the well-being of communities by ensuring that essential medicines are within reach, irrespective of financial constraints.

Problem Solving:

The project addresses the critical problem of limited access to necessary medications. In many cases, individuals are unable to afford essential medicines, leading to prolonged illnesses and compromised health. Cureunity bridges this gap by establishing a user-friendly system that facilitates the free distribution of medicines to those in need, thereby promoting better health outcomes.

Innovation and Enhancement:

Cureunity introduces a novel application that not only provides a new service but also enhances existing healthcare initiatives. By incorporating features such as online medicine ordering, real-time order tracking, and dynamic analysis of medicine sales, the project improves the efficiency and accessibility of healthcare services. Additionally, the inclusion of an 'Add Medicine' tab empowers administrators to dynamically update the database with new medicines, ensuring that the system remains current and comprehensive.

Application/Service Date and Release:

The Cureunity application is set to be released on [insert date], marking a significant milestone in the journey toward equitable healthcare access. This release will empower users to order medicines, track their orders, and benefit from the dynamic analysis of medicine sales. The 'Add Medicine' feature will also be available to administrators, facilitating the continuous expansion and improvement of the medicine database.

In conclusion, Cureunity is not just a project; it's a transformative solution to a critical healthcare challenge. By leveraging technology and innovation, it strives to make a lasting impact on communities by ensuring that essential medicines are accessible to all, regardless of their financial circumstances.

1.1 Competitive Information

In the realm of accessible healthcare solutions, CureUnity enters a landscape with a mix of established and emerging competitors. While our project stands out in terms of its unique features and commitment to social responsibility, it's important to acknowledge existing solutions that address similar needs.

- Competitor Products, Applications, or Services:

Online Pharmacy Platforms: Established online pharmacy platforms allow users to purchase medicines online, but they often involve costs. Unlike CureUnity, these platforms might not specifically cater to individuals facing financial constraints.

Healthcare Apps: Various healthcare applications offer services ranging from appointment scheduling to health record management. However, very few focus on the free distribution of essential medicines, making CureUnity a unique player in this space.

Nonprofit Healthcare Initiatives: Some nonprofit organizations provide free medicines to communities in need. However, these initiatives might lack the technological infrastructure and dynamic features present in CureUnity, such as real-time order tracking and sales analysis.

- Potential to Be First to Market:

CureUnity positions itself as a pioneering solution by combining the accessibility of online pharmacies with the philanthropic mission of nonprofit healthcare initiatives. While established players exist in both realms, the unique combination of features, including real-time order tracking, dynamic sales analysis, and a user-friendly 'Add Medicine' tab for administrators, sets CureUnity apart.

The potential to be the first to market is bolstered by the specific focus on providing free essential medicines, a niche that has not been widely explored in the current healthcare technology landscape. The incorporation of a chat support option on all pages further distinguishes CureUnity by offering personalized assistance to users in real-time.

1.2 Relationship to Other Applications/Projects

CureUnity stands as a standalone innovation with a singular focus on providing free essential medicines to individuals in need. While the project does not involve the management of health information, it is positioned to operate independently, efficiently addressing the critical challenge of limited access to medications.

- Independent Operation:

CureUnity does not require integration with external health information systems, as its primary function revolves around the distribution of medicines. This independence ensures a straightforward user experience, with a streamlined focus on medicine ordering, real-time tracking, and dynamic sales analysis.

- Collaboration with Medicine Suppliers:

While not directly related to health information, Cure Unity may explore collaborations with medicine suppliers to enhance its database and ensure a comprehensive list of available medications. This collaboration could involve updating the platform with new medicines and maintaining accurate stock information.

- Synergy with Community Outreach Programs:

The project can explore synergies with community outreach programs that focus on healthcare awareness and assistance. By aligning with such initiatives, CureUnity can extend its reach and ensure that individuals in need are aware of the free medicine distribution service.

In essence, CureUnity operates as an independent and focused solution, concentrating on addressing the specific challenge of medicine accessibility. While it does not directly relate to health information management, potential collaborations with medicine suppliers and community outreach programs can enhance the project's effectiveness and outreach without compromising its core simplicity and functionality.

1.3 Assumptions and Dependencies

CureUnity operates within a set of assumptions and dependencies to ensure its seamless integration, functionality, and impact. It's crucial to identify these elements for effective project planning and execution.

- **Assumptions:**

Timely Availability of Medicine Information:

Assumption: The availability and accuracy of medicine information from suppliers will be timely and consistent.

Rationale: To maintain an up-to-date and comprehensive medicine database, it is assumed that information provided by suppliers will be accessible in a timely manner.

User Internet Accessibility:

Assumption: Users have internet access to utilize the online platform for ordering and tracking medicines.

Rationale: The project relies on online accessibility for users to place orders, track deliveries, and engage with the platform's features.

Collaboration with Healthcare Professionals:

Assumption: Collaborations with healthcare professionals for prescription validation are feasible.

Rationale: To ensure responsible medicine distribution, validation of prescriptions may be required, assuming collaboration with healthcare professionals is achievable.

- **Dependencies:**

Database Management System:

Dependency: The project relies on the functionality and reliability of the underlying SQL Server database.

Mitigation: Regular maintenance and monitoring of the database to address potential issues and ensure continuous availability.

Internet Service Providers (ISPs):

Dependency: The platform's functionality is dependent on the reliability of internet service providers.

Mitigation: Monitoring and communication plans to address any potential service disruptions, with consideration for users in areas with varying internet accessibility.

1.4 Future Enhancements

CureUnity is envisioned as a dynamic platform that can evolve to meet emerging healthcare needs and technological advancements. Future enhancements are strategically planned to ensure the sustained impact and relevance of the project.
Phase-wise Implementation:

Phase 1 - Initial Release:
Launch the core functionality of CureUnity, focusing on free medicine distribution, real-time order tracking, and dynamic sales analysis.
Establish collaborations with medicine suppliers and community outreach programs to enhance the platform's effectiveness.

Phase 2 - Telemedicine Integration:
Explore integration with telemedicine platforms to create a more interconnected healthcare ecosystem.
Enable users to seamlessly transition from online consultations to ordering and tracking prescribed medicines through CureUnity.

Phase 3 - Geographic Expansion:
Extend the reach of CureUnity to cover a broader geographic area, ensuring that individuals in diverse regions have access to free essential medicines.

1.5 Definitions and Acronyms

Definitions:

- CureUnity:

Definition: The project name, representing a platform dedicated to providing free access to essential medicines.

- MVC Core (Model-View-Controller Core):

Definition: A software architectural pattern that separates an application into three interconnected components: Model (data), View (user interface), and Controller (handling user input).

- SQL Server:

Definition: A relational database management system (RDBMS) developed by Microsoft, used for storing and retrieving data as requested by other software applications.

- Medicine Database:

Definition: A structured collection of data within CureUnity containing information about available medicines, including names, descriptions, and stock levels.

- Real-Time Order Tracking:

Definition: A feature allowing users to monitor the status and location of their medicine orders in real-time, providing transparency and assurance.

- Sales Analysis Dashboard:

Definition: A visual interface within CureUnity that dynamically displays trends and patterns related to medicine sales, enabling users and administrators to make informed decisions.

Acronyms:

- EHR:

Acronym: Electronic Health Record
Definition: A digital version of a patient's paper chart, containing a comprehensive record of the patient's medical history, diagnoses, medications, treatment plans, immunization dates, allergies, radiology images, and laboratory test results.

- ISP:

Acronym: Internet Service Provider
Definition: A company that provides internet access to customers, typically through wired or wireless connections.

- AI:

Acronym: Artificial Intelligence
Definition: The simulation of human intelligence processes by machines, especially computer systems. These processes include learning, reasoning, problem-solving, understanding natural language, and perception.

2 *Project Technical Description*

CureUnity is designed with a focus on simplicity, usability, and scalability. The project utilizes the Model-View-Controller (MVC) Core framework for its architectural foundation, ensuring a modular and maintainable structure.

2.1 *Application Architecture*

CureUnity follows the MVC architectural pattern to separate concerns and enhance modularity:

- Model:

The Model represents the data and business logic of the application.
The Medicine Database, managed by a SQL Server database, serves as the primary model, containing information about available medicines.

- View:

The View is responsible for presenting data to the user and receiving user input.
Various views include the online platform for users, the analysis dashboard, and the administrator interface.

- Controller:

The Controller handles user input, processes requests, and updates the Model and View accordingly.
Controllers manage the flow of data between the Model and View, facilitating user interactions such as ordering medicines and updating the medicine database.

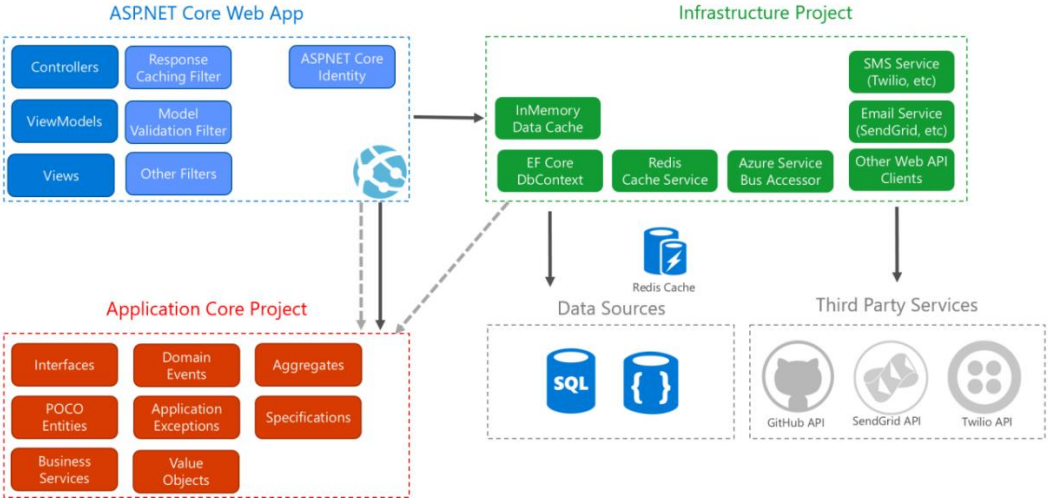

Figure 1 (MVC Architecture)

2.2 Application Information flows

CureUnity is designed to offer a straightforward and user-friendly experience for individuals seeking free essential medicines. The following steps outline the user process from accessing the platform to tracking medicine orders:

1. User Registration:
Users begin by registering on the CureUnity platform, providing necessary details such as name, contact information, and a secure password.We used the identity model for the user registration and authorizations. A secure identity model is paramount for the protection of user data and the prevention of unauthorized access in web applications. Multi-Factor Authentication (MFA) stands as a foundational element, requiring users to provide multiple forms of identification for access, significantly reducing the risk of compromise. Strong encryption protocols, such as HTTPS, safeguard sensitive information during transmission. Robust password policies, encompassing complexity requirements and regular changes, enhance the strength of user credentials. Role-Based Access Control (RBAC) ensures that users only have access to necessary resources, limiting the potential impact of compromised accounts. Continuous monitoring and auditing, facilitated by comprehensive logs, allow for the detection of suspicious activities and rapid response to security incidents. Account lockout policies defend against brute-force attacks, while regular security audits, user education on best practices, and adherence to established security standards collectively contribute to a resilient identity model. Secure account recovery mechanisms, coupled with adherence to industry-recognized principles, further solidify the application's security, creating a dynamic defense against evolving threats.

1. **<u>Login</u>**

```
public async Task<IActionResult> OnPostAsync(string returnUrl = null)
{
    returnUrl ??= Url.Content("~/");

    ExternalLogins = (await _signInManager.GetExternalAuthenticationSchemesAsync()).ToList();

    if (ModelState.IsValid)
    {
        // This doesn't count login failures towards account lockout
        // To enable password failures to trigger account lockout, set lockoutOnFailure: true
        var result = await _signInManager.PasswordSignInAsync(Input.Email, Input.Password, Input.RememberMe, lockoutOnFailure: false);
        if (result.Succeeded)
        {
            _logger.LogInformation("User logged in.");
            var userId = Input.Email;
            HttpContext.Session.SetString("UserId", userId);
            return LocalRedirect(returnUrl);
        }
        if (result.RequiresTwoFactor)
        {
            return RedirectToPage("./LoginWith2fa", new { ReturnUrl = returnUrl, RememberMe = Input.RememberMe });
        }
        if (result.IsLockedOut)
        {
            _logger.LogWarning("User account locked out.");
            return RedirectToPage("./Lockout");
        }
        else
        {
            ModelState.AddModelError(string.Empty, "Invalid login attempt.");
            return Page();
        }
    }

    // If we got this far, something failed, redisplay form
    return Page();
}
No issues found
```

Figure 2(LOGIN_PAGE_Code)

2. **Sign-out**

```
using Microsoft.Extensions.Logging;

namespace Cureunity.Areas.Identity.Pages.Account
{
    public class LogoutModel : PageModel
    {
        private readonly SignInManager<IdentityUser> _signInManager;
        private readonly ILogger<LogoutModel> _logger;

        public LogoutModel(SignInManager<IdentityUser> signInManager, ILogger<LogoutModel> logger)
        {
            _signInManager = signInManager;
            _logger = logger;
        }

        public async Task<IActionResult> OnPost(string returnUrl = null)
        {
            await _signInManager.SignOutAsync();
            _logger.LogInformation("User logged out.");
            if (returnUrl != null)
            {
                return LocalRedirect(returnUrl);
            }
            else
            {
                // This needs to be a redirect so that the browser performs a new
                // request and the identity for the user gets updated.
                return RedirectToPage();
            }
        }
    }
}
```

Figure 3(SIGN_OUT_Code)

2. Browsing Medicines:
After successful registration, users can browse the available medicines on the platform. The Medicine Database provides comprehensive information about each medicine, including names, descriptions, and available stock levels.

3. Placing an Order:
Once users have identified the required medicines, they can place an order through the intuitive online platform. The ordering process includes selecting the desired medicines, specifying quantities, and confirming the order.

4. Order Confirmation:
Users receive a confirmation notification after successfully placing an order. This notification includes an order ID for reference.

5. Real-Time Order Tracking:
Users can track the status and location of their medicine orders in real-time using the dedicated order tracking feature. This provides transparency and assurance throughout the delivery process.

7. Receiving Medicines:
Upon successful order processing and delivery, users receive their essential medicines at the provided delivery address.

8. Providing Feedback:
Users have the option to provide feedback on their experience, enabling continuous improvement of the platform.

9. Accessing Analysis Dashboard (Optional):
Administrators can access the analysis dashboard to track trends in medicine sales dynamically. This feature provides insights into the popularity of different medicines.

```
function gcharts() {
onst textdata = $("#kd").text();

    const cleanedTextData = textdata.replace(/[\r\n]+/g, '');

    // Convert the cleaned text to a JavaScript object
    const data = JSON.parse(cleanedTextData);


    const output = {
        "cols": [
            { "id": "", "label": "Topping", "pattern": "", "type": "string" },
            { "id": "", "label": "Slices", "pattern": "", "type": "number" },
        ],
        "rows": [],
    };

    data.forEach(item => {
        output.rows.push({
            "c": [
                { "v": item.MedicineName, "f": null },
                { "v": parseInt(item.TotalQuantity), "f": null },
            ]
        });
    });

    var data21 = JSON.stringify(output, null, 2);
    var data2 = new google.visualization.DataTable(data21);
    var options = {
        title: 'Medicines Delivered'
    };
    var chart = new google.visualization.BarChart(document.getElementById('chart_div'));
    chart.draw(data2, options);
    var chart = new google.visualization.PieChart(document.getElementById('chart_divpie'));
    chart.draw(data2, options);

}
```

Figure 4(Taking data from SQL to create graphical representation.)

```
function bargraph(dataArray) {
    const dataTable = new google.visualization.DataTable();

    // Add columns to the DataTable
    dataTable.addColumn('string', 'MedicineName');
    dataTable.addColumn('number', 'TotalQuantity');

    // Add data rows to the DataTable from the provided array
   // dataTable.addRows(dataArray);
    dataArray.forEach(item => {
        dataTable.addRows([[item.MedicineName, parseInt(item.TotalQuantity, 10)]]);
    });
    // Set chart options
    const options = {
        title: 'Medicine Data'
    };

    // Draw the chart
    const chart = new google.visualization.BarChart(document.getElementById('myChart'));
    chart.draw(dataTable, options);

}
```

Figure 5 (BAR GRAPH)

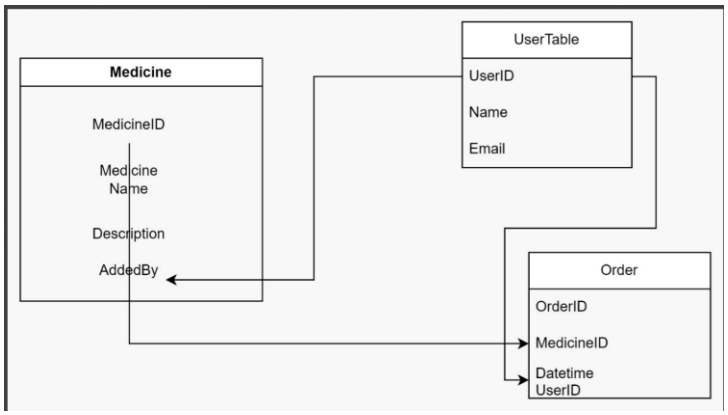

Figure 6 (Class Diagram)

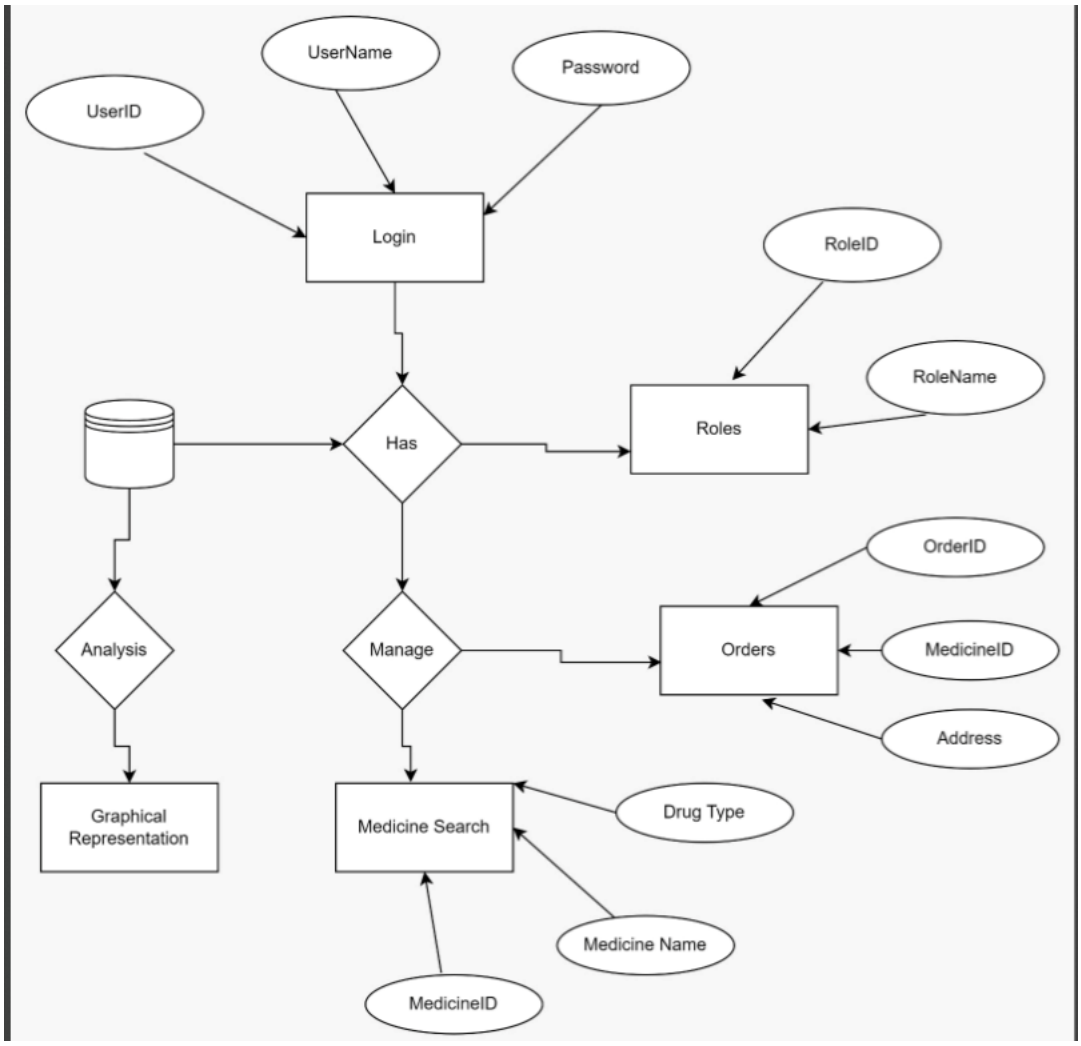

Figure 7 (ER-Diagram)

10. Chat Support (Optional):
At any point in the process, users can utilize the chat support option available on all pages for real-time assistance and guidance. We used the Botpress tool to built the chatbot, Botpress is an open-source conversational platform designed for building and deploying chatbots across various platforms. The platform's RESTful API enables developers to interact programmatically, offering functionalities such as bot and conversation management, user administration, and message handling. Botpress follows a modular and extensible architecture, allowing developers to enhance bot capabilities through modules and middleware, integrating external services and customizing bot behavior.

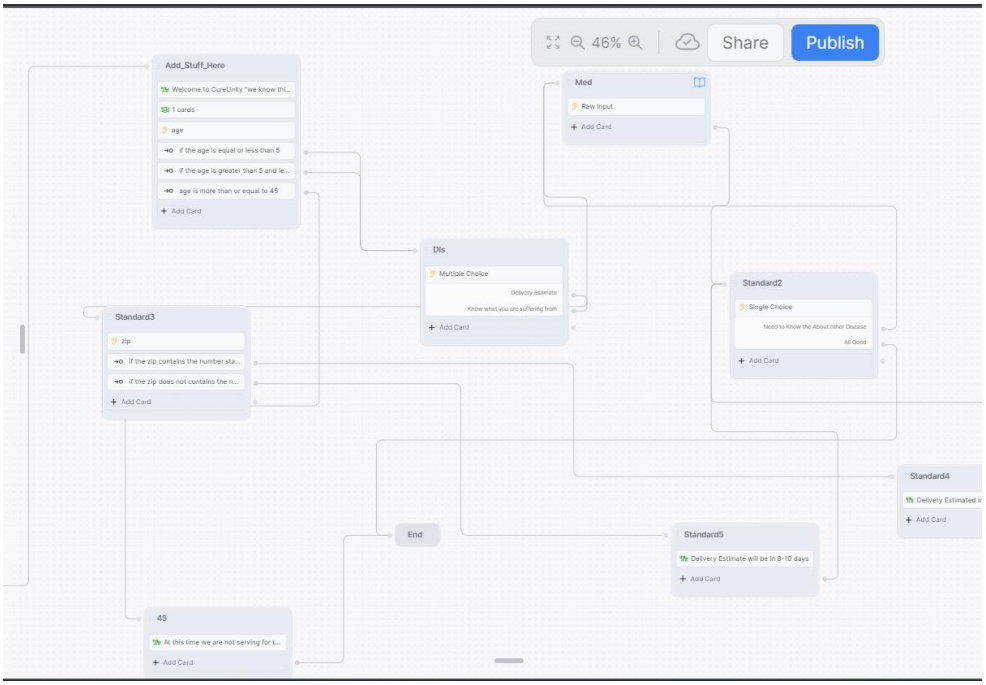

Figure 8_ChatBot architecture

2.3 Interactions with other Projects (if Any)

CureUnity is primarily a standalone project.

2.4 Interactions with other Applications

CureUnity may interact with various external applications to enhance its functionality and provide a more comprehensive user experience. Here are potential interactions and the measures needed to address them:

- **Interaction with External Medicine Information Databases:**

Description: CureUnity may need to interact with external databases that provide additional information about medicines.

Addressing Interaction: Establish API integrations or data exchange protocols to seamlessly retrieve and update information from external medicine databases.

2.5 Capabilities

CureUnity relies on a set of key capabilities to support its core functions and provide a seamless experience for users and administrators. Here is a list of essential capabilities required for the successful implementation and operation of the project:

1. Medicine Ordering System:

Capability: Enable users to browse, select, and order essential medicines through a user-friendly online platform.
Functions:
Retrieve and display information about available medicines from the Medicine Database.
Allow users to add selected medicines to their order.
Facilitate a secure and user-friendly checkout process.

2. Real-Time Order Tracking:

Capability: Provide users with the ability to track the status and location of their medicine orders in real-time.
Functions:
Display the current status of the order (e.g., processing, out for delivery, delivered).
Utilize geolocation services to provide accurate location data during the delivery process.

3. Medicine Database Management:

Capability: Empower administrators to dynamically update and manage the Medicine Database.
Functions:
Provide an 'Add Medicine' tab for administrators to add new medicines to the database.
Enable administrators to update existing medicine information, including stock levels.

4. Sales Analysis Dashboard:

Capability: Offer a dynamic analysis tab for users and administrators to track trends in medicine sales.
Functions:
Display visual representations of sales data, such as graphs and charts.
Allow users and administrators to filter and analyze sales data based on various parameters.

5. Chat Support Integration:

Capability: Provide users with real-time assistance through a chat support option available on all pages.
Functions:
Enable users to initiate and engage in real-time chat sessions with support representatives.
Integrate chat support functionalities seamlessly into the platform.

6. User Registration and Authentication:

Capability: Ensure secure user registration and authentication processes.
Functions:
Collect and validate user registration information.
Authenticate users securely during login.

2.6 Risk Assessment and Management

The successful implementation of CureUnity is contingent on identifying potential risks early in the development process and implementing effective strategies to mitigate or eliminate these risks. Here is a comprehensive risk assessment along with recommended strategies:

1. Data Security and Privacy Concerns:

Risk: Unauthorized access or data breaches may compromise sensitive user information.

Mitigation:

Implement robust encryption protocols for user data.

Conduct regular security audits and vulnerability assessments.

Comply with industry standards and regulations for data protection.

2. Integration Challenges with External Systems:

Risk: Difficulties in integrating with external databases, payment processing systems, or healthcare platforms may hinder functionality.

Mitigation:

Conduct thorough testing during the integration phase.

Establish clear communication and collaboration channels with external system providers.

Develop contingency plans for potential integration issues.

3. Limited Internet Accessibility for Users:

Risk: Users in certain regions may face limited internet accessibility, impacting their ability to use the platform.

Mitigation:

Implement offline functionality where possible.

Provide clear communication about platform requirements and potential limitations.

4. Medicine Supply Chain Disruptions:

Risk: Disruptions in the supply chain may lead to inaccurate information about medicine availability.

Mitigation:

Establish strong collaborations with reliable medicine suppliers.

Implement regular updates and communication with suppliers to ensure accurate stock information.

5. Technical Issues with Real-Time Tracking:

Risk: Technical glitches may affect the accuracy of real-time order tracking.

Mitigation:

Thoroughly test and optimize the real-time tracking system.

Provide alternative communication channels for users to inquire about order status.

6. User Adoption Challenges:

Risk: Users may face difficulties in adopting the platform, affecting its overall success.

Mitigation:

Implement a user-friendly interface with clear instructions.

Conduct user training sessions or provide instructional materials.

Gather user feedback for continuous improvement.

3 Project Requirements

3.1 Identification of Requirements

Requirements for CureUnity are identified and numbered using the SMART criteria, ensuring that each requirement is Specific, Measurable, Achievable, Realistic, and Time-related. The following format is used:

Requirements:

<CureUnity-CU2023-v1-User-Capability-0001>

Specific: Enable user registration and account management.
Measurable: Users should be able to create, update, and delete their accounts.
Achievable: Develop a user-friendly interface for account management.
Realistic: The system must handle user account changes efficiently.
Time-related: Implementation should be completed in the initial release.

<CureUnity-CU2023-v1-Medicine-Capability-0002>

Specific: Implement a comprehensive Medicine Database.
Measurable: Database should contain information about medicine names, descriptions, and stock levels.
Achievable: Collaborate with medicine suppliers to ensure accurate and timely updates.
Realistic: The system must efficiently manage and update the Medicine Database.
Time-related: Database should be fully functional at the initial release.

<CureUnity-CU2023-v1-Order-Capability-0003>

Specific: Facilitate the ordering of essential medicines.
Measurable: Users should be able to select medicines, specify quantities, and complete the ordering process.
Achievable: Develop a secure and user-friendly online ordering platform.
Realistic: The system must handle order processing and communication effectively.
Time-related: Ordering functionality should be available at the initial release.

<CureUnity-CU2023-v1-Tracking-Capability-0004>

Specific: Provide real-time order tracking for users.
Measurable: Users should be able to monitor the status and location of their orders.
Achievable: Implement geolocation services for accurate tracking.
Realistic: The tracking system must be reliable and responsive.
Time-related: Real-time tracking should be functional at the initial release.

<CureUnity-CU2023v1-Analysis-Capability-0005>

Specific: Develop a Sales Analysis Dashboard for users and administrators.
Measurable: Display visual representations of medicine sales trends.
Achievable: Create a dynamic and user-friendly dashboard interface.
Realistic: The system must efficiently analyze and present sales data.
Time-related: Analysis functionality should be available at the initial release.

<CureUnity-CU2023-v1-Chat-Capability-0006>

Specific: Integrate a chat support option available on all pages.
Measurable: Users should be able to initiate and engage in real-time chat sessions with support representatives.
Achievable: Implement WebSocket technology for real-time communication.
Realistic: The chat support feature must be seamlessly integrated.
Time-related: Chat support should be functional at the initial release.

3.2 Operations, Administration, Maintenance and Provisioning (OAM&P)

CureUnity incorporates a set of capabilities and requirements to empower users and administrators in effectively managing and maintaining the platform. This section outlines key aspects related to operations, administration, maintenance, and provisioning:

1. User Account Management:
Capability: Users can manage their accounts, including updates and deletions.
Requirements:
<CureUnity-CU2023-v1-User-OAM&P-0001>
Specific: Users can update their personal information.
Measurable: Changes to user information should reflect accurately.
Achievable: Provide an intuitive user interface for account management.
Realistic: Users should be able to modify non-sensitive account details.
Time-related: Account management should be available in the initial release.

2. Medicine Database Maintenance:
Capability: Administrators can dynamically update and manage the Medicine Database.
Requirements:
<CureUnity-CU2023-v1-Medicine-OAM&P-0002>
Specific: Admins can add new medicines to the database.
Measurable: Newly added medicines should be visible in the platform.
Achievable: Provide an 'Add Medicine' tab for administrators.
Realistic: Admins should be able to manage medicine information efficiently.
Time-related: Database maintenance should be available in the initial release.

3. Order Processing and Tracking Maintenance:
Capability: Ensure smooth order processing and tracking functionalities.
Requirements:
<CureUnity-CU2023-v1-Order-OAM&P-0003>
Specific: Admins can view and manage orders in the system.
Measurable: Order status changes should be reflected accurately.
Achievable: Provide an administrative interface for order management.
Realistic: Admins should be able to handle order-related tasks efficiently.
Time-related: Order management should be available in the initial release.

4. Analysis Dashboard Maintenance:
Capability: Admins can manage and update the Sales Analysis Dashboard.
Requirements:
<CureUnity-CU2023-v1-Analysis-OAM&P-0005>
Specific: Admins can customize and configure dashboard settings.
Measurable: Changes to dashboard configurations should be reflected accurately.
Achievable: Provide an administrative interface for dashboard management.
Realistic: Admins should be able to efficiently update and customize the dashboard.
Time-related: Dashboard management tools should be available in the initial release.

6. Chat Support System Maintenance:
Capability: Admins can monitor and address issues with the chat support system.
Requirements:
<CureUnity-CU2023-v1-Chat-OAM&P-0006>
Specific: Admins can access chat logs and troubleshoot support issues.
Measurable: System logs should provide detailed information on chat interactions.
Achievable: Implement logging mechanisms for chat support functionalities.
Realistic: Admins should have tools to identify and address chat support system issues.
Time-related: Chat support system maintenance tools should be available in the initial release.

3.3 Security and Fraud Prevention

1. User Authentication and Authorization:
Requirement:
<CureUnity-CU2023-v1-Security-0001>
Specific: Implement secure user authentication mechanisms.
Measurable: User credentials must be encrypted during transmission and storage.
Achievable: Utilize industry-standard encryption protocols (e.g., HTTPS).
Realistic: Ensure that user authentication is resistant to common attacks (e.g., brute force).
Time-related: Authentication mechanisms should be in place at the initial release.

2. Data Encryption:
Requirement:
<CureUnity-CU2023-v1-Security-0002>
Specific: Encrypt sensitive user data.
Measurable: Implement end-to-end encryption for user information.
Achievable: Use encryption algorithms to protect data at rest and in transit.
Realistic: Sensitive data should be unreadable even if unauthorized access occurs.
Time-related: Encryption protocols should be implemented at the initial release.

3. Access Control:
Requirement:
<CureUnity-CU2023-v1-Security-0003>
Specific: Enforce access control mechanisms.
Measurable: Users and administrators should have role-based access.
Achievable: Implement role-based access controls to restrict unauthorized access.
Realistic: Users should only have access to functionalities relevant to their roles.
Time-related: Access controls should be functional at the initial release.

4. Medicine Database Integrity:
Requirement:
<CureUnity-CU2023-v1-Security-0004>
Specific: Ensure the integrity of the Medicine Database.
Measurable: Implement data validation to prevent unauthorized modifications.
Achievable: Use checksums or hash functions to verify database integrity.
Realistic: Database integrity checks should be robust and efficient.
Time-related: Database integrity measures should be in place at the initial release.

3.4 Release and Transition Plan

The release and transition plan for CureUnity outlines the process by which the project will be deployed to customers, ensuring a smooth transition from the current release to newer releases. The plan includes key steps and considerations for successful deployment:

1. Deployment Strategy:
Objective: Ensure a seamless transition to new releases while minimizing disruption to users.
Steps:
Staging Environment Testing: Conduct thorough testing of new releases in a staging environment to identify and address any issues before deployment.
Incremental Rollouts: Consider deploying new releases incrementally to specific user groups to monitor and address any unforeseen issues.
Automated Deployment Tools: Utilize automated deployment tools to streamline the process and reduce the likelihood of human errors during deployment.

2. User Communication:
Objective: Keep users informed about upcoming releases, changes, and improvements.
Steps:
Release Notes: Prepare detailed release notes outlining new features, improvements, and any changes in functionality.
Email Notifications: Send proactive email notifications to users about scheduled releases, expected downtime (if any), and post-release instructions.

In-App Announcements: Utilize in-app announcements to communicate changes directly to users upon logging in.

3. Rollback Plan:
Objective: Have a contingency plan in case issues arise during or after deployment.
Steps:
Backup Systems: Ensure that backup systems are in place to facilitate a rollback to the previous release if needed.
Monitoring Tools: Implement monitoring tools to quickly identify and respond to any anomalies or performance issues post-deployment.
Communication Protocol: Define a clear communication protocol for notifying users and stakeholders in the event of a rollback.

4. Training and Documentation:
Objective: Provide training and updated documentation to users and administrators for new features or changes.
Steps:
Training Sessions: Conduct training sessions for administrators on any new capabilities or changes in the administrative interface.
Documentation Updates: Update user manuals, guides, and help documentation to reflect changes in the latest release.
Online Resources: Provide online resources, such as video tutorials or FAQs, to assist users in adapting to new features.

4 *Project Design Description*

1. Architecture Overview:
Description:
CureUnity follows a multi-tier architecture, comprising a presentation layer, application layer, and data layer.
The presentation layer consists of a user-friendly web interface accessible to users and administrators.
The application layer houses the core logic of the platform, handling user interactions, order processing, and administrative functions.
The data layer stores essential information, including user data, medicine details, and order information, in a secure SQL Server database.

2. User Interaction:
Description:
Users interact with the platform through an intuitive web interface.
The interface allows users to register, log in, browse medicines, place orders, and track order status.
Real-time communication channels, including a chat support option, provide immediate assistance to users on any page.

3. Medicine Database Management:
Description:
Administrators access a secure administrative interface to manage the Medicine Database.
The 'Add Medicine' tab enables the addition of new medicines, with fields for name, description, and stock level.
Database updates trigger real-time reflections in the user interface, ensuring accurate and up-to-date information.

4. Order Processing and Tracking:
Description:
Users initiate medicine orders through the web interface.
The order processing system handles the placement of orders, inventory updates, and prescription validation.
Real-time tracking utilizes geolocation services to provide users with live updates on the status and location of their orders.

5. Sales Analysis Dashboard:
Description:
The Sales Analysis Dashboard is accessible to both users and administrators.
Users can view visual representations of medicine sales trends, helping them make informed decisions.
Administrators can analyze data dynamically, filtering sales information based on various parameters.

6. Security Measures:
Description:
Robust security measures, including encrypted user authentication, data encryption, and role-based access controls, ensure the protection of user data.
Access controls extend to external integrations, securing communication with third-party services.
Fraud detection mechanisms, powered by machine learning algorithms, contribute to proactive security.

5 Internal/external Interface Impacts and Specification

1. User Interface (UI) Changes:
Impact:
The web interface will undergo enhancements to accommodate new features such as real-time tracking, chat support, and the Sales Analysis Dashboard.
UI adjustments may include additional buttons, tabs, and interactive elements to support the improved user experience.

2. Administrative Interface Updates:
Impact:
The administrative interface for managing the Medicine Database will introduce a new 'Add Medicine' tab, allowing administrators to seamlessly update the database.
UI changes will accommodate the dynamic nature of the Sales Analysis Dashboard, enabling administrators to customize and configure settings.

3. Real-Time Tracking Integration:
Impact:
A real-time tracking system will be integrated into the user interface to provide live updates on the status and location of medicine orders.
New UI elements, such as maps and order status indicators, will be introduced to convey real-time tracking information to users.

4. Chat Support Integration:
Impact:
A chat support option will be integrated into all pages of the platform, allowing users to initiate and engage in real-time chat sessions with support representatives.
A chat widget or icon will be visible in the corner of all pages, providing easy access to support.

6 Design Units Impacts

CureUnity employs a modular and interconnected design, with a primary focus on a single design unit that covers all requirements. This integrated design ensures a cohesive structure, allowing for seamless collaboration between various functional areas. Below is an overview of the primary design unit and its impact:

Overview:

The primary design unit encapsulates the entire system, including the web interface, application logic, database management, order processing, tracking, analysis, security, and external integrations.
This unit acts as the central hub for all functionalities, fostering a unified and coherent architecture.

6.1 Functional Area A/Design Unit A

User Interface Enhancements:
Impact: Design Unit A will undergo updates to accommodate new features, such as real-time tracking, chat support, and the Sales Analysis Dashboard. Additional buttons, tabs, and interactive elements may be introduced to enhance the overall user experience.

Real-Time Tracking Integration:
Impact: The web interface will incorporate elements for real-time tracking, providing users with live updates on the status and location of their medicine orders. Maps and order status indicators may be introduced to convey tracking information.

Chat Support Integration:
Impact: The chat support option will be seamlessly integrated into all pages of the platform, visible in the corner for easy access. The web interface will provide users with the ability to initiate and engage in real-time chat sessions with support representatives.

Documentation Updates:
Impact: User manuals, guides, and help documentation related to the web interface will be updated to reflect changes in features, functionalities, and the overall user experience. Documentation for external interfaces and APIs will also be revised.

6.1.1 Functional Overview

- Web Interface and User Interaction
  Design Unit A, focusing on the web interface and user interaction, undergoes several enhancements to provide an enriched and intuitive experience for users and administrators. The key functional impacts include:

- User Interface Enhancements:
  Design Unit A will see improvements to the user interface, introducing additional elements to support features like real-time tracking, chat support, and the Sales Analysis Dashboard. This ensures that the platform remains user-friendly and visually appealing.

- Real-Time Tracking Integration:
  The integration of a real-time tracking system within the web interface enables users to receive live updates on the status and location of their medicine orders. This feature enhances transparency and engagement.

- Chat Support Integration:
  The seamless integration of chat support across all pages of the platform allows users to initiate real-time chat sessions with support representatives. This feature provides immediate assistance and improves overall user satisfaction.

6.1.2 Impacts

- User Interface Enhancements:
  Impact: The introduction of real-time tracking, chat support, and the Sales Analysis Dashboard necessitates updates to the user interface. Additional buttons, tabs, and interactive elements will be integrated to accommodate these features, creating a more dynamic and engaging interface.

- Real-Time Tracking Integration:
  Impact: The web interface will be modified to seamlessly integrate real-time tracking elements. Maps, order status indicators, and live updates will be incorporated to provide users with transparent and immediate information about the status and location of their medicine orders.

- Chat Support Integration:
  Impact: The chat support option will be embedded across all pages of the platform, requiring modifications to the web interface for optimal placement and accessibility. This feature ensures users can initiate and engage in real-time chat sessions with support representatives.

6.1.3 Requirements

Design Unit A addresses several project requirements related to the web interface and user interaction. The following requirements are covered by this functional area:

<CureUnity-CU2023-v1-UI-0001>: The web interface must support the addition of new elements for real-time tracking, chat support, and the Sales Analysis Dashboard.
<CureUnity-CU2023-v1-UI-0002>: The user interface should seamlessly integrate real-time tracking elements, including maps and order status indicators, to provide live updates on the status and location of medicine orders.
<CureUnity-CU2023-v1-UI-0003>: The chat support option must be embedded across all pages of the platform, ensuring users can initiate and engage in real-time chat sessions with support representatives.
<CureUnity-CU2023-v1-UI-0004>: Documentation for the web interface, external interfaces, and APIs must be updated to reflect changes in features, functionalities, and overall user experience.

6.2 Functional Area B/Design Unit B

Overview:
Design Unit B is dedicated to the management of the Medicine Database within CureUnity. It includes functionalities related to the addition of new medicines, updates to existing entries, and ensuring accurate and real-time information for users and administrators.

6.2.1 Functional Overview

Addition of New Medicines:
Design Unit B incorporates a user-friendly 'Add Medicine' tab in the administrative interface. This feature empowers administrators to effortlessly add new medicines to the database, providing essential details such as name, description, and stock levels.

Database Updates for Real-Time Reflections:
To ensure accuracy and immediacy, any changes made to the Medicine Database, whether through additions or updates using the 'Add Medicine' tab, trigger real-time reflections in the user interface. This mechanism guarantees that users are presented with the latest and most precise information.

Documentation Updates:
The introduction of the 'Add Medicine' functionality necessitates updates to user manuals, guides, and help documentation related to Medicine Database management. Concurrently, documentation for external interfaces and APIs will be revised to incorporate the changes in functionalities

6.2.2 Impacts

Impacts:

Addition of New Medicines:
- Impact: Design Unit B introduces a new 'Add Medicine' tab in the administrative interface. This feature allows administrators to seamlessly add new medicines to the database, specifying details such as name, description, and stock levels.

Database Updates for Real-Time Reflections:
- Impact: Changes made in the Medicine Database, including additions and updates through the 'Add Medicine' tab, trigger real-time reflections in the user interface. This ensures that users have access to the most accurate and up-to-date information.

Documentation Updates:
- Impact: User manuals, guides, and help documentation related to Medicine Database management will be updated to reflect the introduction of the 'Add Medicine' functionality. Documentation for external interfaces and APIs will also be revised.

6.2.3 Requirements

<CureUnity-CU2023-v1-DB-0001>: The system must provide administrators with the ability to add new medicines to the Medicine Database through a dedicated 'Add Medicine' tab in the administrative interface.

<CureUnity-CU2023-v1-DB-0002>: Changes made to the Medicine Database, including additions and updates through the 'Add Medicine' tab, must trigger real-time reflections in the user interface to ensure users have immediate access to accurate and up-to-date information.

<CureUnity-CU2023-v1-DB-0003>: Documentation for Medicine Database management, including the 'Add Medicine' functionality, must be updated in user manuals, guides, and help documentation. Additionally, documentation for external interfaces and APIs will be revised to reflect changes in functionalities.

7 Open Issues

As of the current draft, there are no open issues. The document has been reviewed and is considered complete for the current stage of development. Any future updates or changes will be addressed in subsequent versions of the document.

8 Acknowledgements

This document was authored by **Saisandesh, Devireddy** with the assistance of **Sashidhar Reddy Devagudi, Phani Verma**. Special thanks to prof. **Xin Chen** for their valuable input and guidance throughout the development of this project design document. Their expertise and feedback have significantly contributed to the clarity and comprehensiveness of the document.

9 References

[1]. De Oliveira, J. (2017). Building Modern Web Apps with ASP.NET Core. [Online] Available at: https://books.google.co.uk/books?hl=en&lr=&id=esxPDwAAQBAJ&oi=fnd&pg=PP1&dq=web+applications+with+ASP.NET+Core.&ots=lZq7Nq7x5Y&sig=_FgYdb4WxTwZ1I5ZjCi7YrOypHo#v=onepage&q=web%20applications%20with%20ASP.NET%20Core.&f=false

[2]. Price, M. J. (2021). .Net6 and c# 10 Modern cross platform development. 6th ed. [Online] Available at: https://books.google.co.uk/books?hl=en&lr=&id=OyNMEAAAQBAJ&oi=fnd&pg=PP1&dq=sql+server+in+mvc+core&ots=cOK57OiqHI&sig=o1JbjoABFgC6UjrZPcMvpT7Xlns#v=onepage&q=sql%20server%20in%20mvc%20core&f=false

[3]. Berardi, N., & Bellinaso, M. (2009). ASP.NET MVC 1.0 Website Programming: Problem Design and Solution. [Online] Available at: https://www.google.co.uk/books/edition/ASP_NET_MVC_1_0_Website_Programming/pTdbqYVM6UQC?hl=en&gbpv=1&dq=integrate+ms+sql+server+to+mvc&printsec=frontcover

10 Appendices

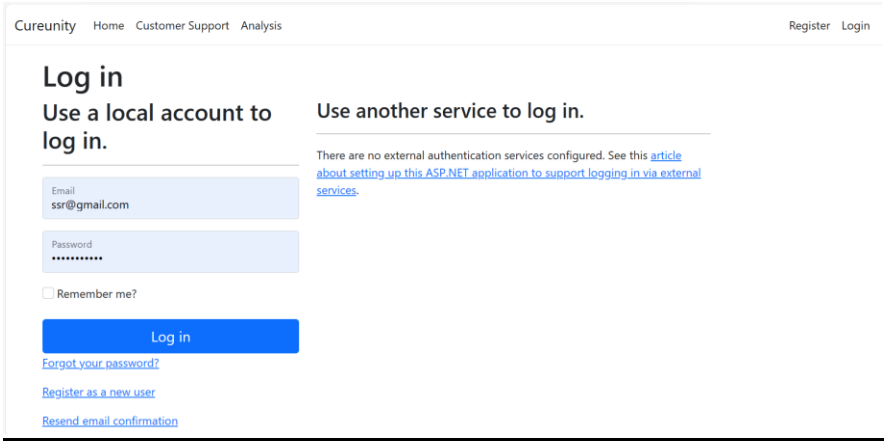

Figure 9(Login Page)

3. Welcome Page

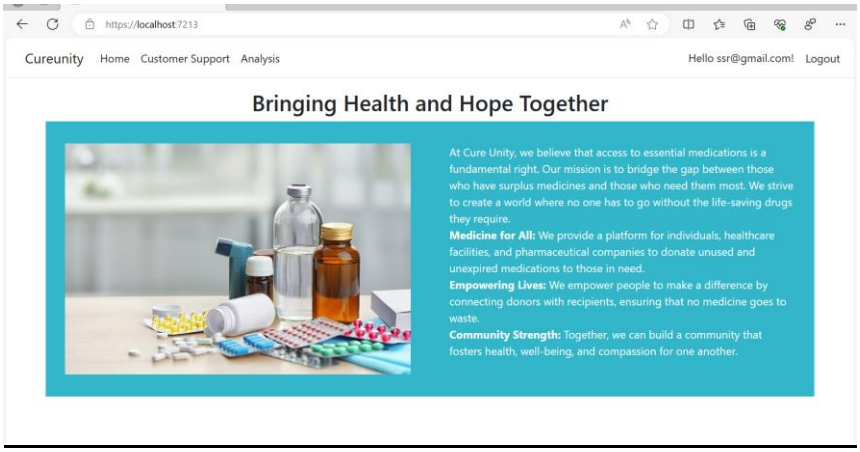

Figure 10 (Welcome Page)

4. Medicine Search

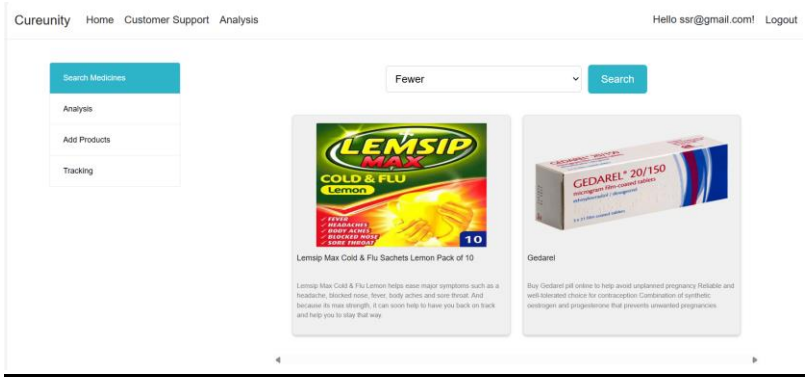

Figure 11 (Medicine Search)

5. Chatbot

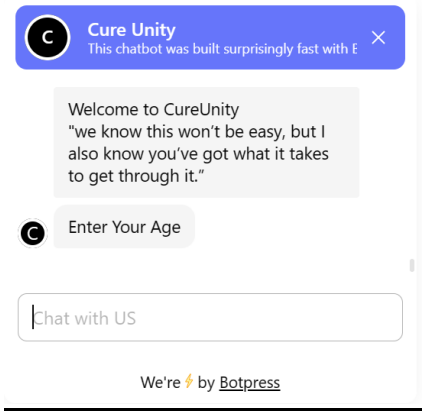

Figure 12 (Chatbot)

6. Analysis

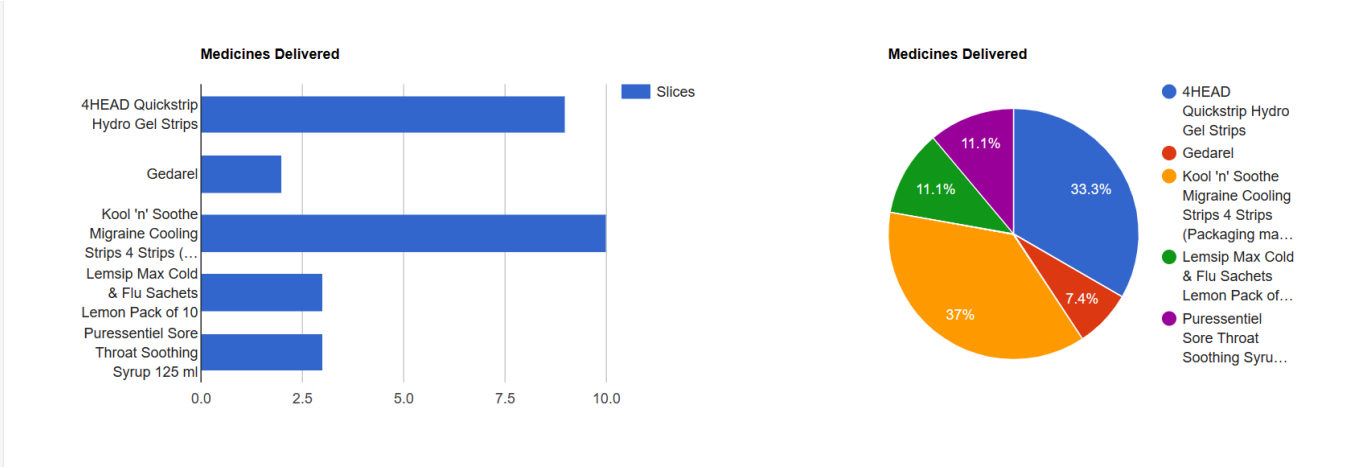

Figure 13 (Analysis Page (Admin))